

Session Objectives

At the conclusion of the session/course, the student should be able to:

Describe the presentation and diagnostic tests used to diagnose the following conditions:

- Rheumatoid Arthritis
- Osteoarthritis
- Gout
- Pseudogout

Describe seronegative spondyloarthritis presentations including: ankylosing spondylitis, psoriatic arthritis, and reactive arthritis.

Differentiate between causes of conditions affecting primarily the joints

Who, What, When, Where, Why, How

Questions to ask as you go through this lecture:

- **What** is it? How does it present, how do you diagnose it?
- **Who** gets it? It occur in certain demographics?
- **When** does it occur? Certain age range?
- **Where** does it occur? Certain joints?
- **How** do we diagnose it? Presentation
- **Why** does it matter? Quality of life?

A 45-year-old woman with joint pain...

A 45 y/o woman presents to her primary care doctor with hand pain and stiffness affecting both hands for the past 6 months...



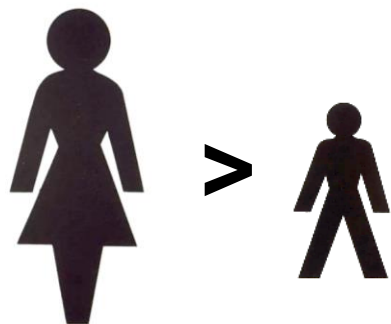
Rheumatoid Arthritis Presentation

What

- Joint pain and stiffness
- Joint swelling and redness
- Symptoms affecting multiple joints including the hands and fingers
- Symmetrical symptoms
- Morning stiffness lasting about 30-60 minutes
 - Or after prolonged inactivity
- Symptoms lasting longer than six consecutive months
- Extra-articular symptoms
 - Malaise, fatigue.

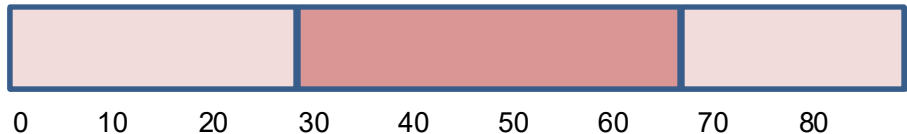
Who

Epidemiology of Rheumatoid Arthritis



Sex:
3:1 female:male

Associated with
Smoking use



Can occur at any age, Peaks 30-40s

When



Affects about 1% of the US
population

Extra-Articular Symptoms/Signs

- Fever, fatigue, weight loss
- Felty Syndrome: RA, Neutropenia, and Splenomegaly
- Rheumatoid nodules

Classic findings of the hands

Swan Neck Deformity



Boutonniere deformity



Joints can have erosions, juxta-articular osteopenia, joint space narrowing, ulnar finger deviation

- Base diagnosis on clinical symptoms and findings...
- Use and radiographic findings to support the diagnosis and rule out similar diseases
- Key labs:
 - Rheumatoid Factor (sensitive but not specific)
 - Anti-Citrullinated peptide Ab (highly specific ~95%)
 - ANA (~30-60% sensitivity)
 - ESR or CRP can help assess severity of dz

RA Treatment

Treat:

- Symptoms, inflammation, and progression

Use DMARDs:

- Corticosteroids, immunosuppressants, methotrexate, biologics: TNF α inhibitors.

- Goals of treatment:
 - Minimize joint pain and swelling,
 - Prevent joint deformity
 - Maintain quality of life
 - Control extra-articular manifestations
 - Both caused by disease and treatment
- Prognosis – dependent on severity however ~12-15 year decrease of life expectancy related to CV disease

Osteoarthritis

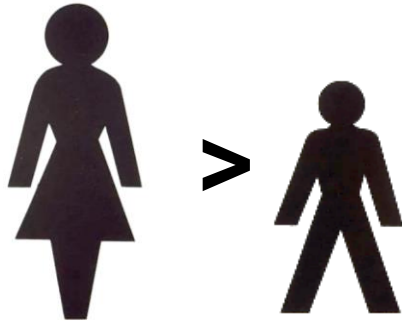
- Degenerative disease of Articular Cartilage
- Secondary Inflammation
- Association:
 - Wear-and-tear
 - Predisposed deformity.

What

Who

Epidemiology of Osteoarthritis

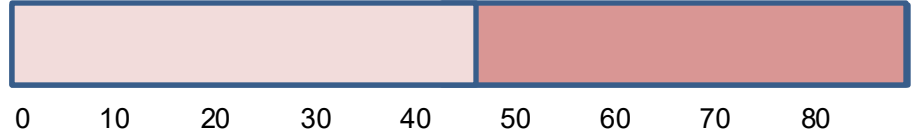
When



Sex:
More Common in
Women than Men



No conclusive
evidence of genetic
relationship



Typically begins in late 40s
with increasing as older



Obesity, competitive contact
athletics, and occupational risks
are positive risk factors

Classic Findings in Osteoarthritis

What

- Insidious onset is often found
- Pain initially is short in duration and later prolonged
- Pain increases with activity or stress
- Joint deformities may further develop and might inhibit activity
- Joints affected are often related to overuse or prior injury (ie, fracture site)

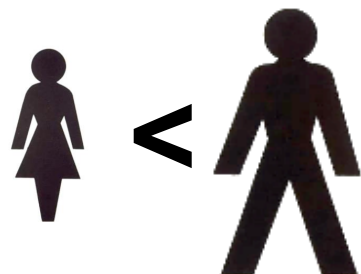
- History and physical
 - Lack of extraarticular manifestations
- Xray findings are indicative
- Laboratory findings will be negative
 - ESR negative
 - Aspiration of synovial fluid should be normal

- Transient Attacks
- Monosodium urate crystals deposits within and around joints.
- Primary cause
 - Enzyme defects
- Secondary causes
 - Renal disease, congenital, leukemia
- Excessive Uric Acid
 - majority is **reduced excretion**
 - Increased rate of synthesis

Who

Epidemiology of Gout

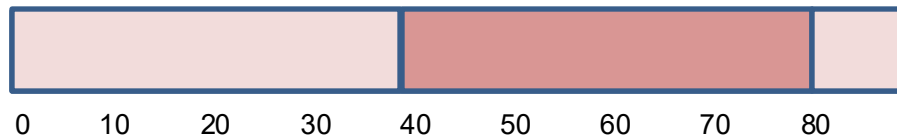
When



Sex:
Female < Male



Possible genetic
role related to
purine metabolism



Can occur at any age however
increased incidence starting in 40s
with a peak at 75.



Variable affect globally related to
diet (higher socioeconomic) and
background (Pacific Islanders)

Classic Presentation of Gout

- Male patient
- Obese
 - or Recent weight gain
- Consumption of EtOH
- Diet high in meat/seafood
- Thiazides
- Renal impairment hx
- Painful joint
- Erythematous
- Swollen
- Warm joint
- Typically monoarticular

Making the diagnosis of Gout

How

- Clinical suspicion
- Joint Aspiration
 - **Negative birefringent crystals**
- Serum Uric Acid ↑
 - 20% of people with high Uric Acid will develop gout



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
Copyright © McGraw-Hill Education. All rights reserved.

Treatment of Gout

Anti-inflammatory

NSAIDS

Colchicine

Glucocorticoids

Urate Lowering Drugs

Uricosuric agents
(probenecid)

Xanthine
Oxidase Inhib
(allopurinol)

Lifestyle

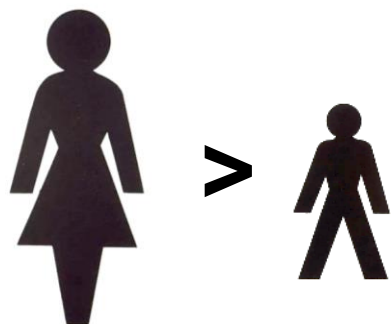
Decrease purine
rich foods

Decreases/stop
EtOh

Weight loss

Who

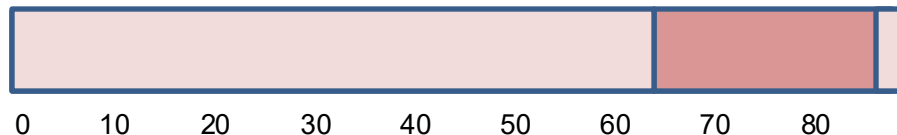
Epidemiology of CPPD Pseudogout



Sex:
Female > Male



Potential genetic
relationship to
pyrophosphate production



Majority of cases after age 50

When

Can be:
Sporadic
Hereditary
secondary

Gout Vs. Pseudogout

- Calcium Pyrophosphate deposition disease
 - Formerly known as Pseudogout
- Crystals form in hyaline cartilage and fibrocartilage.
 - develop in articular cartilage, menisci, and intervertebral discs.
- *Knee: Think pseudogout, not Gout*
- *Toe: think gout*

Where

Making the diagnosis of Pseudogout

- Clinical history
 - Typically, inclusive of a prior injury of the joint
 - Recent reinjury (could be minor)
- Warm, painful joint with fever
- Aspiration of joint with weakly positive birefringent crystals

What

How

Treatment of Pseudogout

- Rest
- Oral or injectable NSAIDs
- Intra-articular glucocorticoid injection
- Recurrent events – consider colchicine prophylaxis
- Surgery if joint destruction/deformity occurs

Seronegative Spondyloarthritis

- Associated with HLA-B27
- Absent Rheumatoid factor.
- Probable T-Cell response.
- Affects ligamentous attachments instead of synovium
- Involves SI and vertebral joints.
- Bony proliferation leading to ankylosis.
- Includes:
 - *Ankylosing spondylitis, Psoriatic Arthritis, Reactive arthritis, and association with Inflammatory Bowel Disease*

Who

Ankylosing Spondylitis



Sex: Historically:
Male > Female



90% HLA-B27 Positive

Starts Second and Third Decade

When

Presentation: lower back pain and
spinal immobility.

1/3 Cases affect peripheral joints.

Reactive Arthritis



Globally:
30/100,000 adults

Who

Assoc. with following infection:
Chlamydia, Shigella, Salmonella, Yersinia, Campylobacter, Clostridioides.

When



HLA-B27

Presentation:

Arthritis

Urethritis/Cervicitis

Conjunctivitis

50% recurrent joint pains